

Daily Geometry Drill #4

Sheet by Henry Koduthore

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Centroid G · Incenter ($A = rs$) · Circumcenter ($R = \frac{abc}{4A}$) · Orthocenter · Angle Bisector ratio · Apollonius medians.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Average the vertices for the centroid, take the hypotenuse midpoint for a right circumcenter, and apply the quick right-triangle inradius rule.

- Write down the centroid of the triangle with vertices $(0, 0)$, $(6, 0)$ and $(3, 9)$.
- The triangle with vertices $(0, 0)$, $(6, 0)$ and $(0, 8)$ is right-angled at the origin. Write down the circumcentre.
- The incircle of the right triangle with legs 3 and 4 touches all three sides. What is the inradius?
- A right triangle with legs 6 and 8 has incircle radius r and circumcircle radius R . Write down $r + R$.
- The sides of a triangle are 6, 8 and 10. Write down its circumradius R .
- Write down the centroid of the triangle with vertices $(1, 2)$, $(5, 6)$ and $(3, -2)$.
- The triangle with vertices $(0, 0)$, $(6, 0)$ and $(0, 8)$ is right-angled. Write down the orthocentre.
- The angle bisector from A of triangle ABC meets BC at D , with $AB : AC = 2 : 3$ and $BC = 10$. Write down DC .
- The isosceles triangle with sides 5, 5, 6 has base 6. Write down the median from the apex to the base.
- A median of a triangle has length 12. How far is the centroid from the vertex at the end of that median?

Section B Manipulation Drills

(≤50 s each)

Feed the side data through $A = rs$, $R = \frac{abc}{4A}$ and the 2:1 median ratio, then check against the options.

- The centroid of the triangle with vertices $(2, 1)$, $(4, 3)$ and $(6, 5)$ is?
A) $(4, 4)$

- B) (3,3)
C) (4,3)
D) (3,4)
2. The incircle of a 5-12-13 right triangle has what radius?
A) 1
B) 2
C) 3
D) 4
3. The triangle with sides 7, 24 and 25 has circumradius?
A) 12
B) $\frac{25}{2}$
C) 25
D) $\frac{37}{2}$
4. The centroid G of a triangle lies on the median AM with $AG = 6$. Find AM .
5. The triangle with vertices $(0,0)$, $(8,0)$ and $(0,6)$ has circumcentre?
A) (3,4)
B) (4,3)
C) (4,4)
D) (6,4)
6. In triangle ABC the angle bisector from A meets BC at D . Given $AB = 6$, $AC = 9$ and $BD = 4$, find DC .
A) 3
B) 6
C) 9
D) $\frac{13}{2}$
7. The isosceles triangle with sides 10, 10, 12 has base 12. The median from the apex to the base is?
A) 6
B) 8
C) 9
D) 10

8. The triangle with sides 13, 14, 15 has area 84. Its inscribed circle has what radius?
9. A right triangle with legs 8 and 15 has inradius?
- A) 2
- B) 3
- C) 4
- D) $\frac{17}{2}$
10. The triangle with sides 13, 14, 15 has area 84. Find its circumradius R (as a fraction).

Section C Substitution & Structure(≤ 75 s each)

Chain two triangle-centre tools per question: locate a centre, then convert through $A = rs$ or $R = \frac{abc}{4A}$ to a second quantity.

1. The centroid of the triangle with vertices $(1, -2)$, $(3, 4)$ and $(-1, 1)$ is?
- A) $(2, 1)$
- B) $(1, 2)$
- C) $(1, 1)$
- D) $(0, 1)$
2. A triangle has area 84 and inradius 4. Its semiperimeter is?
- A) 21
- B) 42
- C) 84
- D) $\frac{21}{2}$
3. The triangle with vertices $(0, 0)$, $(8, 0)$ and $(4, 6)$ has circumcentre?
- A) $\left(4, \frac{5}{3}\right)$
- B) $(4, 3)$
- C) $\left(2, \frac{5}{3}\right)$
- D) $\left(5, \frac{5}{3}\right)$
4. A 3-4-5 triangle has inradius 1 and circumradius $\frac{5}{2}$. What is the distance between its incenter and circumcentre?
- A) $\frac{\sqrt{5}}{2}$

- B) $\frac{3}{2}$
C) 1
D) $\sqrt{2}$
5. The internal angle bisector from A divides BC in the ratio $3 : 4$, and $BC = 14$. What is the length of the shorter segment BD ?
- A) 6
B) 8
C) 3
D) 4
6. The triangle with vertices $(0, 0)$, $(8, 0)$ and $(2, 6)$ has centroid G . What is the distance from G to the origin O ?
- A) $\frac{2\sqrt{34}}{3}$
B) $\frac{10}{3}$
C) 2
D) 4
7. The triangle with vertices $(0, 0)$, $(6, 0)$ and $(3, 4)$ has orthocentre?
- A) $\left(3, \frac{9}{4}\right)$
B) $(3, 4)$
C) $\left(4, \frac{9}{4}\right)$
D) $(3, 3)$
8. For the triangle with vertices $(0, 0)$, $(8, 0)$ and $(4, 6)$, the centroid is $(4, 2)$ and the circumcentre is $\left(4, \frac{5}{3}\right)$. The orthocentre lies on the Euler line three times as far from the centroid as the circumcentre. What is the orthocentre?
- A) $\left(4, \frac{8}{3}\right)$
B) $\left(4, \frac{5}{3}\right)$
C) $(4, 2)$
D) $\left(\frac{8}{3}, 4\right)$

Section D Challenge Ramp(≤ 90 s each)

Stack inradius, circumradius and the bisector/median ratios into multi-step centres.

1. The triangle with vertices $(0,0)$, $(9,0)$ and $(0,12)$ has inradius r and circumradius R . What is $r + R$?
 - A) $\frac{21}{2}$
 - B) 12
 - C) 10
 - D) 15
2. In triangle ABC , $AB = 12$, $AC = 16$ and $BC = 14$. The internal angle bisector from A meets BC at D . What is the length of BD ?
 - A) 6
 - B) 7
 - C) 8
 - D) 9
3. A triangle has sides 6, 7 and 8. What is the length of the median to the longest side (the side of length 8)?
 - A) $\frac{\sqrt{106}}{2}$
 - B) $\frac{\sqrt{53}}{2}$
 - C) $\sqrt{53}$
 - D) $\sqrt{26}$
4. The triangle with vertices $(0,0)$, $(3,0)$ and $(0,4)$ has side lengths 3, 4 and 5. The incenter I is the weighted average of the vertices weighted by the *opposite* side lengths. Find I .
 - A) $(1,1)$
 - B) $(1,2)$
 - C) $(2,1)$
 - D) $\left(\frac{4}{3}, \frac{4}{3}\right)$
5. A 13-14-15 triangle has inradius 4 and circumradius $\frac{65}{8}$. Euler's formula gives the distance d between incenter and circumcentre via $d^2 = R^2 - 2Rr$. What is d ?
 - A) $\frac{\sqrt{65}}{8}$
 - B) $\frac{65}{8}$

C) 4

D) $\frac{1}{2}$

“A triangle is four centres asking the same three sides: average, bisect, and divide by four times the area.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths